RTDL Status Summary (Embree Phase)

Project state

? RTDL is now a real executable prototype: Python-hosted DSL, compiler IR, lowering/codegen, CPU reference runtime, and a working Embree backend on this Mac.

? Current audited precision is float_approx. NVIDIA/OptiX remains the final v0.1 target, but the active baseline is Embree-based execution and evaluation.

? The current working tree also includes Goal 13 paper-reproduction planning, Goal 14 Section 5.6 local-profile planning, and the completed Goal 15 native C++ comparison slice.

Supported workloads

? Six audited workload families are implemented end to end: lsi, pip, overlay, ray_tri_hitcount, segment_polygon_hitcount, and point_nearest_segment.

? The first four use the BVH-oriented Embree path. The last two are currently audited native_loop local workloads, not BVH-accelerated traversal.

? Queries can run through run_cpu(...) for semantic reference and run_embree(...) for native execution.

Reliability process

? Feature and audit rounds are executed with multi-agent review, revision, tests, and archived evidence in history/.

? Goal 12 closed the trust-audit revision under a 2-agent Codex + Gemini rule while Claude was quota-limited.

? Goal 15 added a separate native C++ + Embree versus RTDL + Embree comparison harness for deterministic correctness and host-overhead measurement.

Prepared locally on 2026-03-31 from the RTDL working tree and RayJoin reference artifacts.

Embree Baseline and Evaluation Status

Published baseline

? The Embree baseline goal is complete: workload contracts are frozen, CPU-vs-Embree parity is encoded, and the baseline runner / benchmark / summary pipeline is in place.

? The current checked-in evaluation matrix covers 17 cases across authored, fixture, derived, and synthetic datasets.

? All published Embree evaluation cases pass parity according to the checked-in 2026-03-30 evaluation summary.

Key measured signals

? Largest current local speedup appears in the ray_tri_hitcount synthetic-large case at 165.14x Embree vs CPU on this Mac baseline.

? Current evaluated lsi/pip/overlay cases are still much smaller than the full RayJoin paper campaign and should be treated as baseline validation, not paper-equivalent results.

? The evaluation harness already emits tables, CSVs, SVG figures, a PDF report, and can now be cross-checked against a standalone native C++ comparison slice for lsi and pip.

Goal 13 paper-reproduction status

? Goal 13 is in progress in the repository, but it is temporarily suspended while Goal 14 scopes practical local Section 5.6 runs on this Mac.

? The dataset provenance document now maps RayJoin paper labels County/Zipcode, Block/Water, and LKAF..LKSA / PKAF..PKSA to explicit dataset families and internal names.

? Completed slices already include the Figure 13 / Figure 14 Embree analogue, while remaining work is the Table 3 analogue, Table 4 analogue, and Figure 15 analogue generators.

Current RTDL Embree Evaluation Snapshot

RTDL Embree Evaluation Report

Generated: 2026-03-30T23:47:05
Iterations: 3
Warmup: 1
Host: macOS-26.3-arm64-arm-64bit-Mach-O

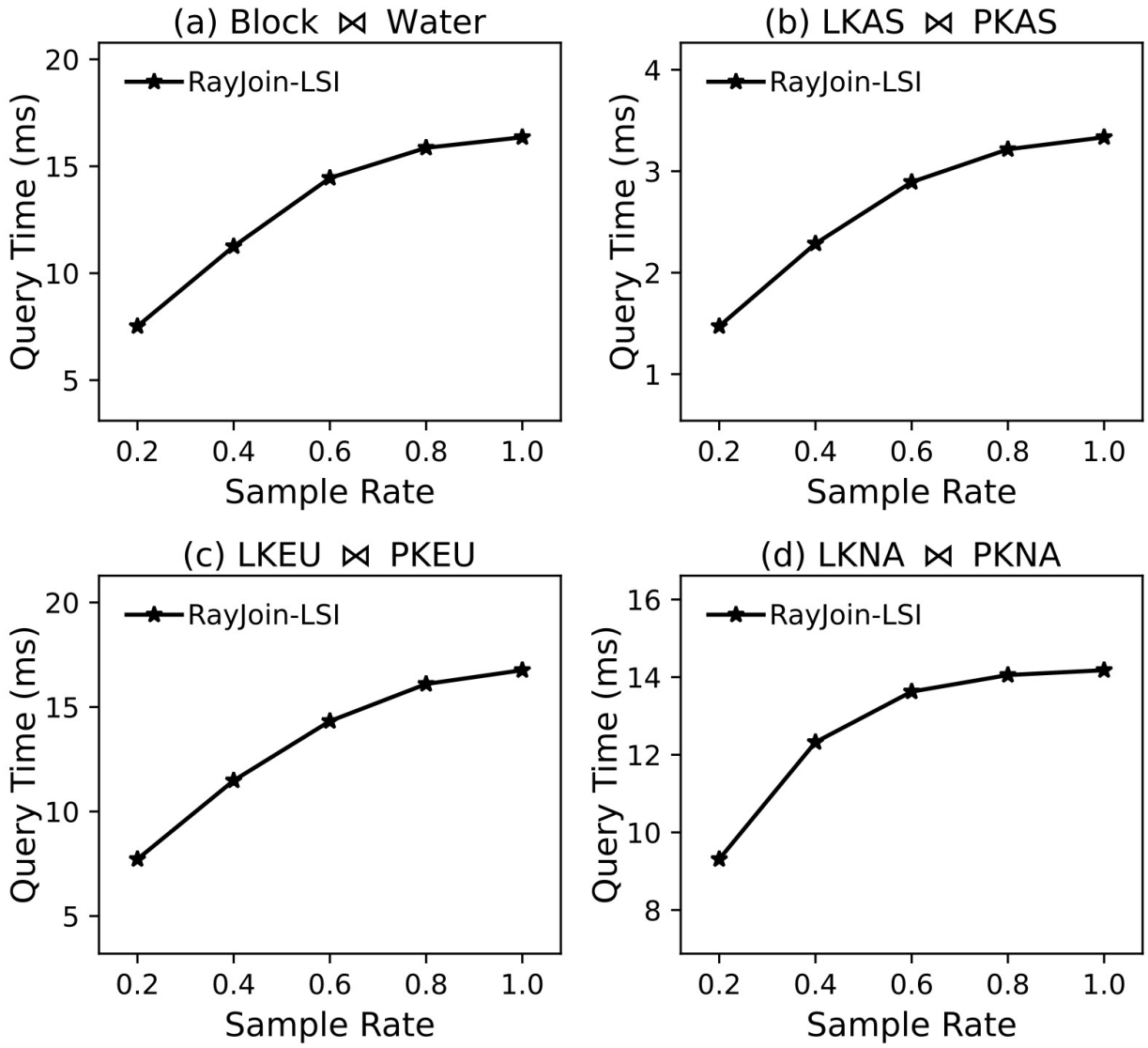
This report summarizes the pre-GPU Embree evaluation for the current RTDL workload surface.
It is an Embree CPU baseline report, not an NVIDIA/RT-core result.

Evaluation Matrix

- lsi_authored_minimal: workload=lsi, category=authored, parity=True, cpu=0.000019s, embree=0.000139s, speedup=0.13x
- lsi_county_slice: workload=lsi, category=fixture, parity=True, cpu=0.000081s, embree=0.000180s, speedup=0.45x
- lsi_county_tiled_x8: workload=lsi, category=derived, parity=True, cpu=0.000704s, embree=0.000413s, speedup=1.70x
- pip_authored_minimal: workload=pip, category=authored, parity=True, cpu=0.000016s, embree=0.000148s, speedup=0.11x
- pip_county_polygons: workload=pip, category=fixture, parity=True, cpu=0.000078s, embree=0.000235s, speedup=0.33x
- pip_county_tiled_x8: workload=pip, category=derived, parity=True, cpu=0.001117s, embree=0.000450s, speedup=2.48x
- overlay_authored_minimal: workload=overlay, category=authored, parity=True, cpu=0.000029s, embree=0.000123s, speedup=0.24x
- overlay_county_soil_fixture: workload=overlay, category=fixture, parity=True, cpu=0.000112s, embree=0.000211s, speedup=0.53x
- overlay_county_soil_tiled_x8: workload=overlay, category=derived, parity=True, cpu=0.000275s, embree=0.000614s, speedup=0.45x
- ray_authored_minimal: workload=ray_tri_hitcount, category=authored, parity=True, cpu=0.000041s, embree=0.000160s, speedup=0.26x
- ray_synthetic_small: workload=ray_tri_hitcount, category=synthetic, parity=True, cpu=0.001173s, embree=0.000223s, speedup=5.27x
- ray_synthetic_medium: workload=ray_tri_hitcount, category=synthetic, parity=True, cpu=0.015852s, embree=0.000437s, speedup=36.24x
- ray_synthetic_large: workload=ray_tri_hitcount, category=synthetic, parity=True, cpu=0.300874s, embree=0.001822s, speedup=165.14x
- segment_polygon_authored_minimal: workload=segment_polygon_hitcount, category=authored, parity=True, cpu=0.000045s, embree=0.000178s, speedup=0.26x
- segment_polygon_county_fixture: workload=segment_polygon_hitcount, category=fixture, parity=True, cpu=0.000339s, embree=0.000294s, speedup=1.15x
- point_nearest_authored_minimal: workload=point_nearest_segment, category=authored, parity=True, cpu=0.000021s, embree=0.000194s, speedup=0.11x
- point_nearest_county_fixture: workload=point_nearest_segment, category=fixture, parity=True, cpu=0.000126s, embree=0.000249s, speedup=0.51x

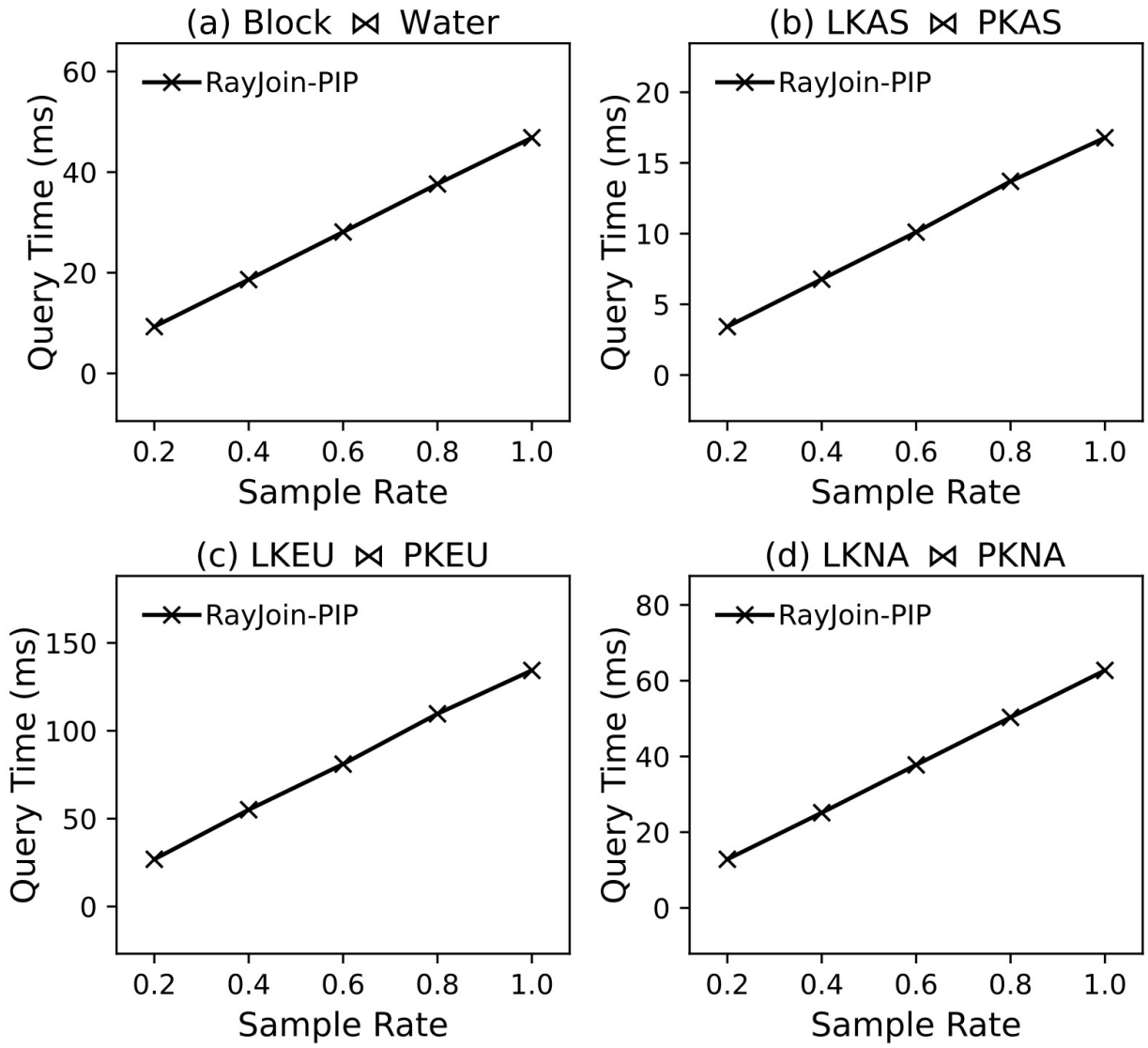
This page is the first-page preview of the current RTDL Embree evaluation report generated inside the repository.
It reflects the existing pre-GPU evaluation pipeline: benchmark cases, parity status, figures, and summary output already work locally.

RayJoin Figure 13 Reference: LSI Scalability



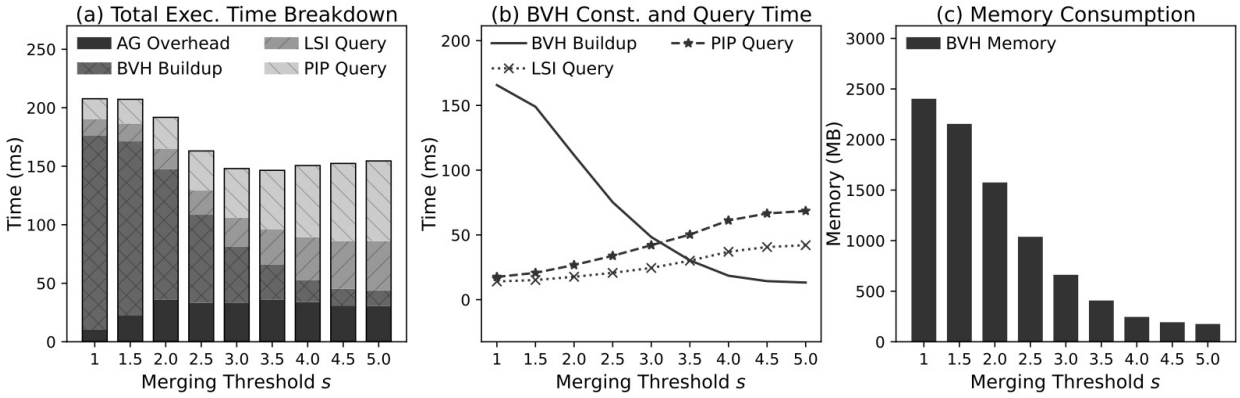
Reference figure from the RayJoin repository. This is the target style for Goal 13 Figure 13 analogues on the Embree phase.
RTDL does not yet regenerate this figure; current local work has only frozen the target matrix, dataset provenance, and registry needed to drive that implementation.

RayJoin Figure 14 Reference: PIP Scalability



Reference figure from the RayJoin repository for PIP scalability.
Goal 13 intends to reproduce the structure, not the GPU numbers: fixed-size synthetic series, uniform and gaussian distributions, query time, and throughput analogues.

RayJoin Figure 15 Reference: Overlay / Varying Enlarge



Reference figure from the RayJoin repository showing overlay-oriented varying-enlarge behavior.
In RTDL, overlay is currently a compositional workload built from lsi and pip seed generation, so the Embree-phase figure will be an analogue and must be labeled honestly.

What is Done vs What Remains

Done now

? Language, docs, examples, CPU runtime, Embree runtime, benchmark harness, history/audit process, and the Embree evaluation report are all real and usable.

? RayJoin paper dataset naming ambiguities for LKAF/LKAS/LKAU/LKEU/LKNA/LKSA have been resolved from the RayJoin experiment scripts.

? A machine-readable paper target registry already exists locally in src/rtdsl/paper_reproduction.py, and Goal 15 now adds a checked-in native C++ comparison report for deterministic lsi/pip fixtures.

Still missing for paper-complete Embree reproduction

? Larger paper-aligned dataset acquisition and conversion, especially County/Zipcode, Block/Water, and continent-scale lakes/parks pairs.

? Automated Table 3 / Table 4 analogue generators and Figure 13 / Figure 14 / Figure 15 analogue generators.

? A final Embree-phase reproduction report that states which paper targets are exact-input, derived-input, fixture-subset, or synthetic-input.

Final v0.1 beyond Embree

? The last major step is the real NVIDIA/OptiX backend so RTDL can rerun RayJoin-style workloads on RT cores and regenerate the paper-style benchmark pipeline with RTDL-generated implementations.